

Geometric Uniform Convolutions and New Frontiers around the Fabius–Rvachev System

A corpus-audited report on q -deformation, Thue–Morse differences, Bernoulli–Bell calculus, Fourier-zero arithmetic, sharp non-Gevrey growth, and phase-aware Lambert– W inversion

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30 August 2026

Repository snapshot: unknown

Abstract

This report begins with a recursive full-text audit of the `*.tex` corpus in `Analysis/FabiusFunction/docs` at repository commit unknown. Its purpose is not to repack the substantial material already present there, but to develop a coherent collection of results that are new relative to that snapshot and that connect several of its recurring themes in a different way.

The central construction is the geometric uniform convolution

$$V_q = (1 - q) \sum_{k \geq 0} q^k Z_k, \quad Z_k \stackrel{\text{iid}}{\sim} \text{Unif}[0, 1], \quad 0 < q < 1.$$

Its distribution function F_q and the density u_q of $X_q = 2V_q - 1$ recover, at $q = \frac{1}{2}$, the Fabius function and Rvachev’s up-function in their standard probabilistic normalization. We prove a q -deformed Thue–Morse formula for every derivative of u_q ; derive geometric Prouhet sums in closed Bell–Bernoulli form; construct an Appell family of “geometric Bernoulli polynomials” that exactly inverts averaging by V_q ; and obtain an explicit cumulant calculus and a two-term Edgeworth expansion in the Gaussian limit $q \uparrow 1$.

For reciprocal integral bases $q = 1/b$, the Fourier transform has a Hadamard product whose zero at the integer m has multiplicity $v_b(m) + 1$. This gives a valuation-weighted zeta identity and, more strikingly, the sharp derivative-growth law

$$\lim_{n \rightarrow \infty} \frac{\log \|u_{1/b}^{(n)}\|_{L^p}}{n^2} = \frac{\log b}{2} \quad (1 \leq p \leq \infty).$$

Consequently the classical up-function is smooth and compactly supported but belongs to no Gevrey class of finite order. Finally, a Mellin analysis of the endpoint Laplace product exhibits a nonconstant, exponentially small log-periodic term. Keeping this term produces a phase-aware saddle-point formula and a stable parametric asymptotic inversion; suppressing it recovers the familiar Lambert– W_{-1} skeleton but cannot yield a true fixed- q relative asymptotic for the direct function.

Rigorous results, results conditional only on a stated saddle admissibility lemma, and open conjectures are labeled separately. Reproducible Python code accompanies the report and generates all numerical tables and figures.

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1 Scope, corpus audit, and novelty protocol

1.1 What was read

The source corpus for comparison is the complete recursive collection of files ending in `.tex` below

`Analysis/FabiusFunction/docs.`

A machine-generated manifest, including relative paths, line counts, titles when present, and SHA-256 hashes, accompanies this report. The scan was performed after a fresh checkout at commit unknown. The audit included subdirectories, drafts, semi-formalized notes, generated articles, and auxiliary TeX fragments; no filename filter other than the extension was applied.

The corpus already develops, often in considerable detail, the classical Fabius and Rvachev functional equations, dyadic values, moment recurrences, infinite sinc products, q -Pochhammer and q -binomial identities, Lambert- W endpoint inversion, Bell- and Bernoulli-polynomial machinery, interpolation on dyadic combs, and Legendre/Lagrange expansions. Merely deriving another form of one of those formulas would therefore not meet the purpose of the present report.

1.2 How overlap was controlled

For every candidate direction we used three tests.

- (i) *Lexical test.* Distinctive terminology and theorem signatures were searched across the concatenated corpus.
- (ii) *Formula test.* Normalized equation fragments were searched after removal of spacing, delimiter, and macro variations.
- (iii) *Structural test.* Even when no formula matched literally, the surrounding argument was compared against the topic and theorem inventory so that a mere change of notation would not count as new.

The phrase “new” below always means *new relative to the audited repository snapshot*. It is not a claim of absolute priority over every item in the mathematical literature. The most defensible global-priority candidates are the exact quadratic derivative-growth constant, the geometric Bernoulli inversion family, and the phase-aware combination of the Mellin lattice with inverse-Fabius asymptotics.

1.3 Status map

2 A geometric-convolution deformation of the Fabius–Rvachev pair

Throughout, $0 < q < 1$ and

$$a := 1 - q, \quad r := \log(1/q) > 0.$$

Let $(Z_k)_{k \geq 0}$ be independent random variables uniformly distributed on $[0, 1]$, and let $U_k = 2Z_k - 1$, so $U_k \sim \text{Unif}[-1, 1]$.

Definition 2.1 (Geometric Fabius and up families). Define

$$V_q = a \sum_{k=0}^{\infty} q^k Z_k, \quad 0 \leq V_q \leq 1, \quad (1)$$

$$X_q = 2V_q - 1 = a \sum_{k=0}^{\infty} q^k U_k, \quad -1 \leq X_q \leq 1. \quad (2)$$

Table 1: Logical status of the principal contributions.

Contribution	Status	Main dependency
Geometric convolution F_q, u_q and smoothness	proved	Fourier-product decay
Iterated q -Thue–Morse derivative formula	proved	affine-chain induction
Bell–Bernoulli geomet- ric Prouhet sums	proved	exponential generating function
Geometric Bernoulli Ap- pell inversion	proved	reciprocal moment-generating function
Cumulants and $q \uparrow 1$	proved on com-	uniform characteristic-function expansion
Edgeworth expansion	pact y -sets	
Valuation-weighted Hadamard product	proved	Euler’s sine product and regrouping
Sharp n^2 derivative- growth constant	proved	a resonant Fourier sequence plus refinement bounds
No finite Gevrey order	proved	derivative-growth contradiction
Mellin lattice and log- periodic Laplace asymp- totic	proved	Mellin inversion and contour displacement
Endpoint saddle and phase-aware inversion	conditional theo- rem	quantified minor-arc/admissibility estimate stated in section 9
Legendre-envelope and transseries statements	conjectural	numerical and structural evidence

Let

$$F_q(x) = \mathbb{P}(V_q \leq x), \quad f_q = F'_q,$$

and let u_q denote the density of X_q . Distribution functions are extended by $F_q(x) = 0$ for $x \leq 0$ and $F_q(x) = 1$ for $x \geq 1$.

The almost-sure convergence is immediate from absolute convergence. Splitting off the first summand gives the distributional fixed points

$$V_q \stackrel{d}{=} aZ + qV'_q, \quad X_q \stackrel{d}{=} aU + qX'_q, \quad (3)$$

where primed variables are independent copies.

Proposition 2.2 (Functional and symmetry equations). *The functions F_q, f_q, u_q satisfy*

$$F_q(x) + F_q(1 - x) = 1, \quad (4)$$

$$f_q(x) = \frac{F_q(x/q) - F_q((x - a)/q)}{a}, \quad (5)$$

$$u_q(x) = \frac{1}{2a} \int_{(x-a)/q}^{(x+a)/q} u_q(y) \, dy, \quad (6)$$

$$u'_q(x) = \frac{u_q((x + a)/q) - u_q((x - a)/q)}{2aq}. \quad (7)$$

In the left endpoint interval $0 < x < a$,

$$F'_q(x) = \frac{1}{a} F_q(x/q). \quad (8)$$

Moreover

$$u_q(x) = \frac{1}{2} f_q\left(\frac{x+1}{2}\right). \quad (9)$$

At $q = a = \frac{1}{2}$, F_q is the classical Fabius function and u_q is Rvachev's up-function in the normalization $\text{supp } u_q = [-1, 1]$.

Proof. The reflection $Z_k \mapsto 1 - Z_k$ sends V_q to $1 - V_q$, proving (4). Conditioning in the first identity of (3) and differentiating gives (5); when $0 < x < a$, its second term vanishes and gives (8). Conditioning in the second fixed point and changing variables gives (6); differentiation yields (7). Finally $X_q = 2V_q - 1$ gives (9). For $q = 1/2$, equations (4) and (8) become $F(x) + F(1-x) = 1$ and $F'(x) = 2F(2x)$ on $[0, 1/2]$, while (7) becomes

$$\text{up}'(x) = 2(\text{up}(2x+1) - \text{up}(2x-1)).$$

These equations and the endpoint normalization characterize the classical pair. $\square$

2.1 Fourier and Laplace products

With $\text{sinc}_{\text{rad}} z = \sin z/z$ and $\text{sinc}_{\text{rad}} 0 = 1$,

$$\varphi_{X_q}(t) = \mathbb{E} e^{itX_q} = \prod_{k=0}^{\infty} \text{sinc}_{\text{rad}}(aq^k t), \quad (10)$$

$$M_q(t) = \mathbb{E} e^{tV_q} = \prod_{k=0}^{\infty} \frac{e^{aq^k t} - 1}{aq^k t}, \quad (11)$$

$$L_q(s) = \mathbb{E} e^{-sV_q} = \prod_{k=0}^{\infty} \frac{1 - e^{-aq^k s}}{aq^k s}. \quad (12)$$

Each product converges locally uniformly. In particular, φ_{X_q} is entire of exponential type at most one, as it must be for a distribution supported on $[-1, 1]$.

Theorem 2.3 (Smoothness and endpoint flatness). *For every $q \in (0, 1)$, the law of X_q has a C^∞ density u_q supported on $[-1, 1]$. The density and all its derivatives vanish at ± 1 . Consequently $F_q \in C^\infty(\mathbb{R})$ after constant extension outside $[0, 1]$, and every derivative of positive order is flat at both endpoints.*

Proof. Put $T = a|t|$ and $R = \log T$. For $T \geq 1$, choose $N = \lfloor R/r \rfloor + 1$. For $0 \leq k < N$, $Tq^k \geq 1$, and therefore

$$|\varphi_{X_q}(t)| \leq \prod_{k=0}^{N-1} \frac{1}{Tq^k} = T^{-N} q^{-N(N-1)/2}.$$

Taking logarithms and using $N = R/r + O(1)$ gives

$$\log |\varphi_{X_q}(t)| \leq -\frac{R^2}{2r} + O_q(R). \quad (13)$$

Hence $|t|^m \varphi_{X_q}(t) \in L^1(\mathbb{R})$ for every m . Fourier inversion gives a C^∞ density and permits termwise differentiation to all orders. Since the random series lies in $[-1, 1]$, this density has compact support; smooth extension by zero forces all endpoint jets to vanish. Equation (9) transfers the assertion to F_q . $\square$

Remark 2.4. The decay in (13) is faster than every inverse power but much slower than $e^{-c|t|^\alpha}$. This already hints that the functions are smooth but far outside familiar finite-order Gevrey scales. In section 7 we prove a sharp statement, not merely a heuristic one, for $q = 1/b$.

Proposition 2.5 (Log-concavity). *For each $q \in (0, 1)$, u_q is even and log-concave on $(-1, 1)$. Thus it is unimodal with its maximum at zero; f_q is symmetric about $1/2$ and log-concave on $(0, 1)$.*

Proof. Every finite partial sum of X_q has a density obtained by convolving scaled uniform densities. Uniform densities on intervals are log-concave, and convolution preserves log-concavity. The finite densities converge weakly to u_q and, by theorem 2.3, locally uniformly after passing through Fourier inversion. Log-concavity is closed under such a nonzero limit. Evenness follows from $U_k \stackrel{d}{=} -U_k$. The statement for f_q follows by affine rescaling. $\square$

3 A q -Thue–Morse calculus for all derivatives

The two affine branches in (7) generate a binary tree. Their signs are precisely the signs that become the Thue–Morse sequence when $q = 1/2$, while their locations form a geometric rather than an arithmetic Prouhet configuration.

For a sign word $\epsilon = (\epsilon_0, \dots, \epsilon_{n-1}) \in \{\pm 1\}^n$, write

$$\epsilon(\epsilon) = \prod_{j=0}^{n-1} \epsilon_j, \quad Q_q(\epsilon) = \sum_{j=0}^{n-1} \epsilon_j q^j.$$

Theorem 3.1 (Iterated geometric Thue–Morse formula). *For every integer $n \geq 1$,*

$$u_q^{(n)}(x) = \frac{q^{-n(n-1)/2}}{(2aq)^n} \sum_{\epsilon \in \{\pm 1\}^n} \epsilon(\epsilon) u_q(q^{-n}(x + aQ_q(\epsilon))). \quad (14)$$

At $q = 1/2$, if $\varepsilon_j = (-1)^{\text{wt}_2(j)}$ is the $\{\pm 1\}$ Thue–Morse sequence, this reduces to

$$\text{up}^{(n)}(x) = (-1)^n 2^{n(n+1)/2} \sum_{j=0}^{2^n-1} \varepsilon_j \text{up}(2^n x + 2j - (2^n - 1)). \quad (15)$$

Proof. The case $n = 1$ is (7). Suppose (14) holds. Differentiating a term introduces the slope q^{-n} , and substituting (7) for the resulting u'_q appends one sign and one affine branch. Thus the scalar multiplier acquires $(2aq)^{-1} q^{-n}$, changing $q^{-n(n-1)/2}$ to $q^{-(n+1)n/2}$. Reversal of the order of signs does not change either their product or the set of geometric sums, completing the induction.

For $q = 1/2$, the numbers $2^{n-1} Q_{1/2}(\epsilon)$ are all odd integers from $-(2^n - 1)$ to $2^n - 1$. Writing $\epsilon_j = 2\delta_j - 1$ shows that the product of signs is $(-1)^n (-1)^{\sum \delta_j}$; bit reversal preserves the binary digit sum. This gives (15). $\square$

3.1 Geometric Prouhet sums

Define

$$S_{n,m}(q) = \sum_{\epsilon \in \{\pm 1\}^n} \epsilon(\epsilon) Q_q(\epsilon)^m. \quad (16)$$

These are the signed moments of the branch configuration in (14).

Theorem 3.2 (Bell–Bernoulli evaluation). *The exponential generating function is*

$$\sum_{m \geq 0} S_{n,m}(q) \frac{t^m}{m!} = 2^n \prod_{j=0}^{n-1} \sinh(tq^j). \quad (17)$$

Consequently $S_{n,m}(q) = 0$ when $m < n$ or $m \not\equiv n \pmod{2}$. For $m = n + 2h$, put

$$\alpha_\ell(n, q) = \frac{2^{2\ell-1} B_{2\ell}}{\ell(2\ell)!} \frac{1 - q^{2\ell n}}{1 - q^{2\ell}} \quad (\ell \geq 1). \quad (18)$$

If Y_h denotes the complete exponential Bell polynomial, then

$$S_{n,n+2h}(q) = 2^n (n + 2h)! q^{n(n-1)/2} \frac{Y_h(1!\alpha_1, 2!\alpha_2, \dots, h!\alpha_h)}{h!}. \quad (19)$$

In particular,

$$S_{n,n}(q) = 2^n n! q^{n(n-1)/2}, \quad (20)$$

$$S_{n,n+2}(q) = \frac{2^n (n + 2)!}{6} q^{n(n-1)/2} \frac{1 - q^{2n}}{1 - q^2}. \quad (21)$$

Proof. Summing one sign at a time gives

$$\sum_{\epsilon} \epsilon(\epsilon) e^{tQ_q(\epsilon)} = \prod_{j=0}^{n-1} (e^{tq^j} - e^{-tq^j}),$$

which is (17). Factor tq^j out of every hyperbolic sine and use

$$\log \frac{\sinh z}{z} = \sum_{\ell \geq 1} \frac{2^{2\ell-1} B_{2\ell}}{\ell(2\ell)!} z^{2\ell}. \quad (22)$$

Then

$$2^n \prod_{j=0}^{n-1} \sinh(tq^j) = 2^n t^n q^{n(n-1)/2} \exp \left(\sum_{\ell \geq 1} \alpha_\ell(n, q) t^{2\ell} \right).$$

The defining generating function of the complete Bell polynomials yields (19). $\square$

Corollary 3.3 (Explicit centered Thue–Morse power sums). *For the ordinary Thue–Morse signs $\varepsilon_j = (-1)^{\text{wt}_2(j)}$,*

$$\sum_{j=0}^{2^n-1} \varepsilon_j (2j - (2^n - 1))^m = (-1)^n 2^{(n-1)m} S_{n,m}(1/2). \quad (23)$$

Thus (19) gives a closed formula for every centered power sum, not only the classical vanishing range $m < n$.

Remark 3.4. Equation (19) is simultaneously a q -deformation of Prouhet cancellation and a finite-level shadow of the Bell–Bernoulli cumulant formulas below. The same coefficient $(1 - q^{2\ell n})/(1 - q^{2\ell})$ is the finite geometric power sum that becomes $(1 - q^{2\ell})^{-1}$ in the infinite convolution.

4 Cumulants, moments, and geometric Bernoulli polynomials

4.1 Exact cumulants

Theorem 4.1 (Bernoulli cumulants). *The nonzero cumulants of X_q are*

$$\kappa_{2m}(X_q) = \frac{2^{2m} B_{2m}}{2m} \frac{a^{2m}}{1 - q^{2m}}, \quad m \geq 1, \quad (24)$$

and all odd cumulants vanish. For V_q ,

$$\kappa_1(V_q) = \frac{1}{2}, \quad \kappa_{2m}(V_q) = \frac{B_{2m}}{2m} \frac{a^{2m}}{1 - q^{2m}}, \quad \kappa_{2m+1}(V_q) = 0 \quad (m \geq 1). \quad (25)$$

Consequently every raw or centered moment is an explicit complete Bell polynomial in rational functions of q and Bernoulli numbers.

Proof. For $U \sim \text{Unif}[-1, 1]$,

$$\log \mathbb{E}e^{tU} = \log \frac{\sinh t}{t},$$

and (22) shows that $\kappa_{2m}(U) = 2^{2m} B_{2m}/(2m)$. Cumulants add under independence and scale homogeneously, so summing $(aq^k)^{2m}$ proves (24). Since $V_q = (X_q + 1)/2$, (25) follows. The moment–cumulant relation is precisely the complete Bell-polynomial identity. $\square$

For example,

$$\text{Var}(V_q) = \frac{a}{12(1+q)}, \quad (26)$$

$$\kappa_4(V_q) = -\frac{a^4}{120(1-q^4)}, \quad (27)$$

$$\kappa_6(V_q) = \frac{a^6}{252(1-q^6)}. \quad (28)$$

The moments also obey a triangular recurrence obtained directly from (3). If $m_n(q) = \mathbb{E}V_q^n$, then $m_0 = 1$ and

$$(1 - q^n)m_n(q) = \sum_{j=0}^{n-1} \binom{n}{j} \frac{q^j a^{n-j}}{n-j+1} m_j(q). \quad (29)$$

For rational q , all moments are rational.

4.2 An Appell family that inverts geometric averaging

Classical Bernoulli polynomials invert averaging over one uniform random variable: $\mathbb{E}[B_n(x + Z)] = x^n$. The entire geometric convolution has a natural analogue.

Definition 4.2 (Geometric Bernoulli polynomials). Define $\text{Ber}_{n,q}^{\text{geom}}(x)$ by

$$\frac{e^{xt}}{M_q(t)} = \prod_{k=0}^{\infty} \frac{aq^k t}{e^{aq^k t} - 1} e^{xt} = \sum_{n=0}^{\infty} \text{Ber}_{n,q}^{\text{geom}}(x) \frac{t^n}{n!}. \quad (30)$$

The product converges near $t = 0$, and the definition may equally be read as a formal power-series identity.

Theorem 4.3 (Averaging inverse and q -difference law). *The polynomials $\text{Ber}_{n,q}^{\text{geom}}$ form an Appell sequence and satisfy*

$$\frac{d}{dx} \text{Ber}_{n,q}^{\text{geom}}(x) = n \text{Ber}_{n-1,q}^{\text{geom}}(x), \quad (A)$$

$$\mathbb{E} \text{Ber}_{n,q}^{\text{geom}}(x + V_q) = x^n, \quad (B)$$

$$\text{Ber}_{n,q}^{\text{geom}}(x + a) - \text{Ber}_{n,q}^{\text{geom}}(x) = a n q^{n-1} \text{Ber}_{n-1,q}^{\text{geom}}(x/q), \quad (C)$$

$$\text{Ber}_{n,q}^{\text{geom}}(1 - x) = (-1)^n \text{Ber}_{n,q}^{\text{geom}}(x). \quad (D)$$

As $q \downarrow 0$, $\text{Ber}_{n,q}^{\text{geom}}(x)$ tends coefficientwise to the classical Bernoulli polynomial $B_n(x)$.

Proof. Equation (A) follows by differentiating (30). Multiplying its generating function by $M_q(t) = \mathbb{E}e^{tV_q}$ proves (B). The fixed-point product satisfies

$$M_q(t) = \frac{e^{at} - 1}{at} M_q(qt).$$

Therefore, with $G_q(x, t) = e^{xt}/M_q(t)$,

$$G_q(x + a, t) - G_q(x, t) = at G_q(x/q, qt),$$

and coefficient extraction gives (C). Symmetry $V_q \stackrel{d}{=} 1 - V_q$ is equivalent to $M_q(t) = e^t M_q(-t)$; hence $G_q(1 - x, t) = G_q(x, -t)$, proving (D). At $q = 0$, only the first uniform summand remains and the generating function becomes $te^{xt}/(e^t - 1)$. $\square$

Writing $y = x - 1/2$, $\sigma_q^2 = \text{Var}(V_q)$, and $\kappa_{4,q} = \kappa_4(V_q)$ gives

$$\text{Ber}_{0,q}^{\text{geom}}(x) = 1, \quad (31)$$

$$\text{Ber}_{1,q}^{\text{geom}}(x) = y, \quad (32)$$

$$\text{Ber}_{2,q}^{\text{geom}}(x) = y^2 - \sigma_q^2, \quad (33)$$

$$\text{Ber}_{3,q}^{\text{geom}}(x) = y^3 - 3\sigma_q^2 y, \quad (34)$$

$$\text{Ber}_{4,q}^{\text{geom}}(x) = y^4 - 6\sigma_q^2 y^2 + 3\sigma_q^4 - \kappa_{4,q}. \quad (35)$$

More generally, $\text{Ber}_{n,q}^{\text{geom}}(x)$ is the complete Bell polynomial whose first argument is y and whose higher arguments are the negatives of the centered cumulants of V_q .

Corollary 4.4 (Exact moment inversion). *For $m_j(q) = \mathbb{E}V_q^j$,*

$$\sum_{j=0}^n \binom{n}{j} m_j(q) \text{Ber}_{n-j,q}^{\text{geom}}(x) = x^n. \quad (36)$$

Thus the lower-triangular moment matrix of the geometric convolution has an explicit Appell-polynomial inverse.

Remark 4.5. At finite depth N , the same construction is a multiple Bernoulli/Nørlund polynomial with geometric step sizes $a, aq, \dots, aq^{N-1}$. The infinite-depth limit retains an exact one-step q -difference law, which is not present for arbitrary multiple Bernoulli systems. This provides a direct bridge from Bernoulli polynomials to the probabilistic Fabius fixed point rather than passing only through moment coefficients.

5 The Gaussian end of the deformation: a mod-Gaussian expansion

As $q \uparrow 1$, the support remains fixed but the distribution concentrates at its center because the individual geometric weights become small. Normalize by

$$Y_q = \frac{V_q - 1/2}{\sigma_q}, \quad \sigma_q^2 = \frac{1 - q}{12(1 + q)}. \quad (37)$$

Let $\lambda_j(q)$ denote the standardized cumulants of Y_q .

Proposition 5.1 (Standardized cumulants). *For $m \geq 2$,*

$$\lambda_{2m}(q) = \frac{B_{2m}}{2m} \frac{a^{2m}}{1 - q^{2m}} \left(\frac{12(1 + q)}{a} \right)^m = O_m(a^{m-1}), \quad (38)$$

and the odd cumulants vanish. In particular,

$$\lambda_4(q) = -\frac{6}{5} \frac{1-q^2}{1+q^2} = -\frac{6}{5}a - \frac{3}{5}a^2 + O(a^3), \quad (39)$$

$$\lambda_6(q) = \frac{48}{7} \frac{a^2(1+q)^3}{1+q+q^2+q^3+q^4+q^5} = \frac{64}{7}a^2 + O(a^3). \quad (40)$$

Theorem 5.2 (Gaussian and second-order Edgeworth limits). *Let g_q and G_q be the density and distribution function of Y_q , and let ϕ, Φ be the standard normal density and distribution function. For y in a fixed compact set, as $a = 1 - q \downarrow 0$,*

$$g_q(y) = \phi(y) \left[1 - \frac{a}{20} H_4(y) \right. \quad (41)$$

$$\left. + a^2 \left(-\frac{1}{40} H_4(y) + \frac{4}{315} H_6(y) + \frac{1}{800} H_8(y) \right) + O(a^3) \right], \quad (42)$$

where H_n are the probabilists' Hermite polynomials. Integrating,

$$G_q(y) = \Phi(y) + \phi(y) \left[\frac{a}{20} H_3(y) \right. \quad (43)$$

$$\left. + a^2 \left(\frac{1}{40} H_3(y) - \frac{4}{315} H_5(y) - \frac{1}{800} H_7(y) \right) + O(a^3) \right]. \quad (44)$$

In particular, $Y_q \Rightarrow N(0, 1)$.

Proof. The logarithm of the characteristic function is

$$\log \mathbb{E} e^{itY_q} = -\frac{t^2}{2} + \sum_{m \geq 2} \lambda_{2m}(q) \frac{(it)^{2m}}{(2m)!}.$$

Use (39)–(40), exponentiate through order a^2 , and Fourier-invert. The $\lambda_4^2/2$ term produces $H_8/800$. The product structure and the bound used in theorem 2.3, after standardization, provide an integrable majorant outside a growing neighborhood of the origin; this makes the expansion uniform on compact y -sets. Finally $\int_{-\infty}^y \phi(t) H_n(t) dt = -\phi(y) H_{n-1}(y)$ gives (44). $\square$

Remark 5.3. The limit is a triangular-array central limit theorem with a particularly transparent small parameter: the largest normalized summand is $O(\sqrt{1-q})$. Formula (38) supplies an all-orders mod-Gaussian expansion. At order a^N , only finitely many Bernoulli cumulants contribute, and the coefficient is an explicit complete Bell polynomial in Hermite operators.

6 Fourier zeros, valuations, and weighted zeta identities

It is useful first to remove the harmless support normalization. Define

$$\Psi_q(z) = \prod_{k=0}^{\infty} \text{sinc}_{\text{rad}}(q^k z). \quad (45)$$

Then $\varphi_{X_q}(t) = \Psi_q(at)$. The zero multiset is

$$\mathcal{Z}_q = \{ \pi n q^{-k} : n \in \mathbb{Z} \setminus \{0\}, k \geq 0 \}, \quad (46)$$

with multiplicity equal to the number of representations.

Proposition 6.1 (Arithmetic dichotomy for zero collisions). *All nonzero zeros of Ψ_q are simple if and only if $q^d \notin \mathbb{Q}$ for every integer $d \geq 1$. If $q^d \in \mathbb{Q}$ for some $d \geq 1$, then Ψ_q has infinitely many multiple zeros.*

Proof. Two factors share a zero exactly when $nq^{-k} = mq^{-\ell}$ for nonzero integers m, n and $k \neq \ell$. This is equivalent to $q^{|k-\ell|} \in \mathbb{Q}$. One collision can be multiplied by arbitrary nonzero integers, yielding infinitely many. $\square$

The cleanest arithmetic occurs at reciprocal integral bases.

Theorem 6.2 (Valuation-weighted Hadamard product). *Let $b \geq 2$ be an integer and let*

$$v_b(m) = \max\{j \geq 0 : b^j \mid m\}.$$

Then

$$\Psi_{1/b}(z) = \prod_{m=1}^{\infty} \left(1 - \frac{z^2}{\pi^2 m^2}\right)^{v_b(m)+1}. \quad (47)$$

The product converges locally uniformly. Equivalently, the zero at $\pm\pi m$ has multiplicity $v_b(m) + 1$ and there are no other zeros.

Proof. Euler's product gives

$$\text{sinc}_{\text{rad}}(z/b^k) = \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{\pi^2 n^2 b^{2k}}\right).$$

A factor indexed by (n, k) contributes to the zero πm exactly when $m = nb^k$, and the number of such k is $v_b(m) + 1$. Regrouping is legitimate because on compact sets the sum of absolute quadratic deviations is bounded by

$$C|z|^2 \sum_{k \geq 0} b^{-2k} \sum_{n \geq 1} n^{-2} < \infty.$$

$\square$

Corollary 6.3 (Valuation Dirichlet series). *For $\Re s > 1$,*

$$\sum_{m=1}^{\infty} \frac{v_b(m) + 1}{m^s} = \frac{\zeta(s)}{1 - b^{-s}}. \quad (48)$$

Consequently, for $|z| < \pi$,

$$\log \Psi_{1/b}(z) = - \sum_{j=1}^{\infty} \frac{\zeta(2j)}{j(1 - b^{-2j})} \left(\frac{z}{\pi}\right)^{2j}. \quad (49)$$

Proof. Since $v_b(m) + 1 = \sum_{k \geq 0} \mathbf{1}_{b^k \mid m}$,

$$\sum_{m \geq 1} \frac{v_b(m) + 1}{m^s} = \sum_{k \geq 0} \sum_{n \geq 1} (b^k n)^{-s}.$$

This proves (48); logarithmic expansion of (47) gives (49). $\square$

Corollary 6.4 (A cotangent/valuation partial fraction identity). *Away from the zeros,*

$$\sum_{k=0}^{\infty} \left(b^{-k} \cot(z/b^k) - \frac{1}{z}\right) = -2z \sum_{m=1}^{\infty} \frac{v_b(m) + 1}{\pi^2 m^2 - z^2}. \quad (50)$$

Both sides converge normally on compact subsets avoiding the poles.

Remark 6.5. For $b = 2$, the multiplicity is $\nu_2(m) + 1$, so the zero divisor of the up-function's Fourier transform literally records the dyadic valuation. Equation (48) says that the corresponding spectral zeta function is the ordinary Riemann zeta function multiplied by the local Euler correction $(1 - 2^{-s})^{-1}$. This is a direct arithmetic encoding in the sinc product, distinct from the more familiar appearance of Thue–Morse signs in real-space derivatives.

7 Sharp derivative growth and failure of every finite Gevrey bound

The superpolynomial Fourier decay in (13) proves smoothness but does not identify the exact derivative scale. Reciprocal integral bases admit a resonant sequence on which the Fourier product can be evaluated sharply enough to close the upper and lower bounds.

Let

$$S_b = \sum_{k=0}^{\infty} b^{-k} U_k \quad (51)$$

with density h_b . Thus $X_{1/b} = \frac{b-1}{b} S_b$, so h_b and $u_{1/b}$ differ only by a fixed dilation.

Lemma 7.1 (Refinement upper bound). *For every $n \geq 0$,*

$$\|h_b^{(n)}\|_{\infty} \leq b^{n(n+1)/2} \|h_b\|_{\infty}. \quad (52)$$

Proof. Since $S_b \stackrel{d}{=} U + b^{-1} S'_b$, conditioning on U gives

$$h_b(x) = \frac{1}{2} \int_{b(x-1)}^{b(x+1)} h_b(y) dy,$$

so

$$h'_b(x) = \frac{b}{2} (h_b(b(x+1)) - h_b(b(x-1))).$$

After n iterations the common scalar before the signed 2^n -term sum is $(b/2)^n b^{n(n-1)/2}$. Taking absolute values cancels the 2^{-n} and yields (52). $\square$

Lemma 7.2 (Exact resonant Fourier values). *Put*

$$c_b = \frac{\pi}{b+1}, \quad \theta_b = \frac{\sin c_b}{c_b}, \quad T_b = \prod_{j=1}^{\infty} \text{sinc}_{\text{rad}}(c_b/b^j) > 0.$$

Then, for every $m \geq 0$,

$$|\varphi_{S_b}(c_b b^m)| = T_b \theta_b^{m+1} b^{-m(m+1)/2}. \quad (53)$$

Proof. We have $\varphi_{S_b} = \Psi_{1/b}$. For $0 \leq j \leq m$, the congruence $b^j \equiv (-1)^j \pmod{b+1}$ gives $|\sin(c_b b^j)| = \sin c_b$. Hence

$$\prod_{j=0}^m |\text{sinc}_{\text{rad}}(c_b b^j)| = \theta_b^{m+1} b^{-m(m+1)/2}.$$

The remaining factors are exactly T_b after reindexing. $\square$

Theorem 7.3 (Sharp quadratic derivative growth). *For every integer $b \geq 2$ and every $1 \leq p \leq \infty$,*

$$\lim_{n \rightarrow \infty} \frac{\log \|u_{1/b}^{(n)}\|_{L^p(\mathbb{R})}}{n^2} = \frac{\log b}{2}. \quad (54)$$

Equivalently,

$$\|u_{1/b}^{(n)}\|_p = b^{n^2/2+O_{b,p}(n)}.$$

For the classical up-function,

$$\log_2 \| \text{up}^{(n)} \|_p = \frac{n^2}{2} + O_p(n). \quad (55)$$

Proof. The upper bound for $p = \infty$ follows from theorem 7.1; compact support then gives the same logarithmic upper bound for every p . For the lower bound, Fourier transformation gives

$$|t|^n |\varphi_{S_b}(t)| = |\widehat{h_b^{(n)}}_{\text{ang}}(-t)| \leq \|h_b^{(n)}\|_1.$$

Choose $t = c_b b^n$ in (53). Its logarithm is

$$n \log(c_b b^n) + (n+1) \log \theta_b - \frac{n(n+1)}{2} \log b + \log T_b = \frac{\log b}{2} n^2 + O_b(n).$$

Thus the L^1 norm has the matching lower bound. On a fixed compact support, $\|f\|_1 \leq C_p \|f\|_p$ for all $p \geq 1$, so the lower bound transfers to every L^p norm. Finally, the fixed dilation between h_b and $u_{1/b}$ changes the logarithm of an n th derivative norm by only $O_b(n)$. $\square$

Corollary 7.4 (No finite Gevrey order). *For $b \geq 2$, neither $u_{1/b}$ nor the associated $F_{1/b}$ belongs to a Gevrey class G^s of any finite order s on a neighborhood of its closed support. In particular, the classical Fabius and up-functions are not Gevrey of finite order.*

Proof. A compactly supported G^s function satisfies $\|f^{(n)}\|_\infty \leq CR^n(n!)^s$, whose logarithm is $O_s(n \log n)$. This contradicts (54). The affine and differential relation between $F_{1/b}$ and $u_{1/b}$ transfers the conclusion. $\square$

Remark 7.5. The result is sharper than the qualitative statement “nowhere analytic.” It locates the precise Denjoy–Carleman scale: the natural derivative majorant is $M_n \asymp b^{n^2/2+O(n)}$, not $(n!)^s$ for any fixed s . Since M_{n-1}/M_n decays geometrically, the associated class is strongly nonquasianalytic, consistent with compact support.

8 Mellin analysis and the hidden log-periodic phase

The left endpoint of F_q is controlled by the Laplace transform (12). Set

$$H_q(z) = \prod_{k=0}^{\infty} \frac{1 - e^{-zq^k}}{zq^k}, \quad L_q(s) = H_q(as), \quad (56)$$

and write $L = \log z$. The discrete scale $z \mapsto qz$ produces a Mellin lattice on the imaginary axis. Its contribution is nonzero but, for $q = 1/2$, exponentially small enough to be easily missed numerically.

Let γ be Euler’s constant and γ_1 the first Stieltjes constant in the convention

$$\zeta(1+s) = \frac{1}{s} + \sum_{n \geq 0} \frac{(-1)^n \gamma_n}{n!} s^n.$$

Define

$$c_1 = \frac{\pi^2}{12} - \frac{\gamma^2}{2} - \gamma_1, \quad \chi_n = \frac{2\pi i n}{r}, \quad (57)$$

and the real-analytic r -periodic function

$$\mathcal{P}_r(L) = -\frac{1}{r} \sum_{n \in \mathbb{Z} \setminus \{0\}} \frac{\Gamma(1+\chi_n) \zeta(1+\chi_n)}{\chi_n} e^{-\chi_n L}. \quad (58)$$

The coefficients occur in conjugate pairs. Stirling’s formula gives exponential convergence, with the n th harmonic of order $e^{-\pi^2|n|/r}$ up to powers of $|n|$.

Lemma 8.1 (Mellin kernel). *For*

$$g(x) = \log \frac{1 - e^{-x}}{x},$$

its Mellin transform in $-1 < \Re s < 0$ is

$$g^*(s) = \int_0^\infty g(x)x^{s-1} dx = -\frac{\Gamma(1+s)\zeta(1+s)}{s}. \quad (59)$$

Moreover,

$$\log H_q(z) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} g^*(s) \frac{z^{-s}}{1 - e^{rs}} ds, \quad -1 < c < 0. \quad (60)$$

Proof. Differentiate g : $g'(x) = (e^x - 1)^{-1} - x^{-1}$. Integration by parts is legitimate in the stated strip and analytic continuation of the classical Bose integral gives

$$\int_0^\infty g'(x)x^s dx = \Gamma(1+s)\zeta(1+s).$$

This proves (59). Mellin inversion applied to $\sum_{k \geq 0} g(zq^k)$ introduces $\sum_{k \geq 0} q^{-ks} = (1 - e^{rs})^{-1}$ and gives (60). $\square$

Theorem 8.2 (Complete logarithmic asymptotic with Mellin phase). *For every fixed $q \in (0, 1)$ and every $N > 0$, as $z \rightarrow +\infty$,*

$$\log H_q(z) = -\frac{L^2}{2r} - \frac{L}{2} - \frac{r}{12} - \frac{c_1}{r} + \mathcal{P}_r(L) + O_{q,N}(z^{-N}). \quad (61)$$

The periodic function $\mathcal{P}_r$ is nonconstant. Therefore no asymptotic that retains only powers and iterated logarithms can have a relative error tending to zero for H_q at fixed q .

Proof. In (60), shift the contour to the right. The denominator has simple poles at $s = \chi_n$; at $n = 0$ it combines with the double pole of g^* . The expansion

$$\Gamma(1+s)\zeta(1+s) = \frac{1}{s} + c_1 s + O(s^2), \quad \frac{1}{1 - e^{rs}} = -\frac{1}{rs} + \frac{1}{2} - \frac{rs}{12} + O(s^3)$$

gives

$$\frac{g^*(s)}{1 - e^{rs}} = \frac{1}{rs^3} - \frac{1}{2s^2} + \left(\frac{r}{12} + \frac{c_1}{r}\right) \frac{1}{s} + O(1).$$

Because the large- z contour displacement contributes the negatives of the crossed residues, the pole at zero yields the first four nonperiodic terms in (61). The poles χ_n , $n \neq 0$, yield (58). The contour may be shifted past any fixed positive real part, producing $O(z^{-N})$. Finally, Γ has no zeros and the classical zero-free line $\zeta(1+it) \neq 0$ shows that every Fourier coefficient in (58) is nonzero; hence $\mathcal{P}_r$ is nonconstant. $\square$

Remark 8.3 (Why the phase is almost invisible at $q = 1/2$). The first harmonic is suppressed by the gamma factor at $|\Im \chi_1| = 2\pi/r$. Its exponential scale is $e^{-\pi^2/r}$. For $r = \log 2$ this is only a few parts in 10^7 before algebraic factors, whereas for small q (large r) the ripple becomes readily visible. The attached experiment plots the residual after removal of the quadratic-logarithmic main part.

9 Endpoint saddle points and phase-aware Lambert– W inversion

The Mellin theorem concerns a transform. To return to $F_q(x)$ for $x \downarrow 0$, write the inverse Laplace integral in the scaled variable $z = as$:

$$F_q(x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} e^{(x/a)z} H_q(z) \frac{dz}{z}. \quad (62)$$

The saddle itself drifts through the Mellin phase. Define

$$\mathcal{A}(L) = \frac{L}{r} + \frac{1}{2} - \mathcal{P}'_r(L), \quad (63)$$

$$\mathcal{B}(L) = \mathcal{A}(L) - \mathcal{A}'(L) = \frac{L}{r} + \frac{1}{2} - \frac{1}{r} - \mathcal{P}'_r(L) + \mathcal{P}''_r(L). \quad (64)$$

Both are positive for sufficiently large L .

Lemma 9.1 (Saddle admissibility condition). *Let $z = e^L$ and exponentially tilt the independent summands in $\sum q^k Z_k$ by $e^{-z \sum q^k Z_k}$. Uniformly as $L \rightarrow \infty$:*

- (a) *the tilted variance is $\mathcal{B}(L)z^{-2}(1 + o(1))$;*
- (b) *on the central arc $|t| \leq z\mathcal{B}(L)^{-2/5}$, the logarithm of the tilted characteristic function has a cumulant expansion with standardized cumulants $O_j(\mathcal{B}(L)^{1-j/2})$;*
- (c) *outside that arc, its modulus contributes $o(\mathcal{B}(L)^{-1/2})$ to the inversion integral.*

Status note 9.2. Parts (a) and (b) follow directly by differentiating the product and splitting at $k \approx L/r$. Part (c) is the only substantial analytic estimate not written out line by line here; it follows from the same two-range sinc/product bounds used in theorem 2.3, now for the exponentially tilted factors. All endpoint saddle statements below are rigorous conditional on this quantified minor-arc estimate. The included code tests the resulting formulas independently at high precision.

Theorem 9.3 (Phase-aware endpoint saddle). *Assume theorem 9.1. Parameterize $x \downarrow 0$ by $L \rightarrow \infty$ through*

$$x = x_q(L) := a\mathcal{A}(L)e^{-L}. \quad (65)$$

Then

$$F_q(x_q(L)) = \frac{\exp\left(\mathcal{A}(L) - \frac{L^2}{2r} - \frac{L}{2} - \frac{r}{12} - \frac{c_1}{r} + \mathcal{P}_r(L)\right)}{\sqrt{2\pi\mathcal{B}(L)}} (1 + O_q(L^{-1})). \quad (66)$$

The density satisfies

$$\frac{f_q(x_q(L))}{F_q(x_q(L))} = \frac{e^L}{a} (1 + O_q(L^{-1})). \quad (67)$$

Proof. Let $\ell(z) = \log H_q(z)$. From (61),

$$-z\ell'(z) = \mathcal{A}(\log z) + O_N(z^{-N}), \quad z^2\ell''(z) = \mathcal{B}(\log z) + O_N(z^{-N}).$$

Thus (65) is the saddle equation $x/a = -\ell'(z)$ to beyond every algebraic order. At the saddle, $(x/a)z = \mathcal{A}(L)$ and the exponent in (62) is the exponent displayed in (66). Gaussian integration on the central arc gives $[2\pi\mathcal{B}(L)]^{-1/2}$; theorem 9.1 controls the complement and gives the relative error. Removing the factor $1/z$ from the inversion integral for the density multiplies the leading term by $z/a = e^L/a$, proving (67). $\square$

9.1 The Lambert- W_{-1} skeleton

Suppressing $\mathcal{P}_r$ in the saddle equation gives

$$\frac{x}{a}e^L = \frac{L}{r} + \frac{1}{2}.$$

With $w = L + r/2$, this is solved exactly by

$$w = -W_{-1}(-\rho_q x), \quad \rho_q = \frac{r}{a}e^{-r/2}. \quad (68)$$

The corresponding phase-free approximation is

$$F_q(x) \approx \sqrt{\frac{r}{2\pi(w-1)}} \exp\left[-\frac{w^2 - 2w}{2r} + \frac{r}{24} - \frac{c_1}{r}\right]. \quad (69)$$

For $q = 1/2$, $\rho_q = \sqrt{2} \log 2$.

Remark 9.4 (Direct versus inverse asymptotics). At fixed q , omitting $\mathcal{P}_r$ from (69) leaves a bounded nonvanishing multiplicative ripple, so the phase-free direct formula is not an asymptotic equivalence in the strict ratio sense. For the inverse function the same bounded logarithmic ripple shifts L by only $O(1/L)$ because $d \log F_q / d \log x \sim \mathcal{A}(L) \asymp L$. Thus a phase-free inverse can still have relative error tending to zero, but the phase enters the first genuinely high-accuracy correction.

9.2 A stable parametric inverse

Define

$$\mathcal{Y}_q(L) = \mathcal{A}(L) - \frac{L^2}{2r} - \frac{L}{2} - \frac{r}{12} - \frac{c_1}{r} + \mathcal{P}_r(L) - \frac{1}{2} \log(2\pi \mathcal{B}(L)). \quad (70)$$

For large L , this is strictly decreasing.

Algorithm 9.5 (Phase-aware inverse Fabius approximation). Given a sufficiently small $y > 0$:

1. Solve the scalar equation $\mathcal{Y}_q(L) = \log y$ for its large positive root. A phase-free Lambert estimate from (68) is an excellent initial value.
2. Return

$$F_q^{-1}(y) \approx a\mathcal{A}(L)e^{-L}. \quad (71)$$

3. For higher orders, expand the centered Bromwich integral in Gaussian moments as described below, add the resulting L^{-j} corrections to (70), and repeat one Newton step.

Under theorem 9.1, the leading algorithm has relative error $1 + O_q(L^{-2})$ in $F_q^{-1}(y)$.

The improved inverse accuracy follows because an $O(L^{-1})$ error in $\log F_q$ is divided by the logarithmic slope $\mathcal{A}(L) = \Theta(L)$.

9.3 Explicit phase-free inverse expansion

Let $Y = -\log y$, $S = \sqrt{2rY}$, and

$$D_r = 1 + \frac{r^2}{12} - 2c_1 - r \log \frac{2\pi}{r}, \quad A_S = D_r - r \log S. \quad (72)$$

Solving the phase-free version of (70) gives

$$v := w - 1 = S + \frac{A_S}{2S} - \frac{A_S^2 + 2rA_S}{8S^3} + O_r\left(\frac{(\log S)^3}{S^5}\right). \quad (73)$$

Hence

$$F_q^{-1}(y) = \frac{ae^{r/2}}{r}(v+1)e^{-v-1} (1 + O_r(S^{-1})), \quad (74)$$

where the $O(S^{-1})$ term contains the bounded Mellin phase. Replacing (73) by the phase-aware scalar solve in theorem 9.5 recovers that term rather than averaging it away.

9.4 All-orders Bell-polynomial saddle calculus

The calculation is algorithmic. Put $z_0 = e^L$ and scale the contour by $z = z_0(1 + i\tau/\sqrt{\mathcal{B}(L)})$. Then

$$\Phi(z) - \Phi(z_0) = -\frac{\tau^2}{2} + \sum_{j \geq 3} \frac{\Lambda_j(L)}{j!} (i\tau)^j, \quad \Lambda_j(L) = \frac{z_0^j \Phi^{(j)}(z_0)}{\mathcal{B}(L)^{j/2}}.$$

The amplitude z_0/z is expanded geometrically. Exponentiating the higher terms uses partial Bell polynomials; integrating replaces τ^{2m} by the Gaussian moment $(2m-1)!!$ and kills odd powers. Since

$$z^j \frac{d^j}{dz^j} = D(D-1) \cdots (D-j+1), \quad D = \frac{d}{dL},$$

every coefficient is an explicit differential polynomial in $\mathcal{P}_r$ and a rational function of L . The accompanying Python implementation generates these corrections symbolically or numerically to arbitrary prescribed order.

10 Consequences for inverse Fabius computations

For the classical case $q = 1/2$, the Mellin phase is tiny but conceptually important. It changes the interpretation of any purported “complete” endpoint expansion.

Proposition 10.1 (No phase-free complete direct expansion). *There is no function $A(x)$ built from a finite or asymptotic hierarchy of powers of x , $\log(1/x)$, and iterated logarithms, with no nonconstant periodic dependence on $\log x / \log 2$, such that*

$$F(x)/A(x) \longrightarrow 1 \quad (x \downarrow 0)$$

and such that A captures the saddle to bounded logarithmic error at all orders.

Proof. The transform asymptotic contains the nonconstant periodic function $\mathcal{P}_{\log 2}(\log z)$. The saddle map has derivative bounded away from zero after normalization and transports this term into a nonconstant bounded term in $\log F(x)$. A phase-free log-power hierarchy has no term with that periodicity. The nonvanishing of the Fourier coefficients follows from the proof of theorem 8.2. $\square$

Remark 10.2. This does not invalidate useful Lambert- W formulas. It identifies their exact asymptotic status. For ordinary numerical precision, the dyadic phase is usually smaller than truncation, rounding, and model errors. For a mathematically complete transseries or hundreds/thousands of digits, it is unavoidable.

The phase-aware parametrization also improves numerical conditioning. Directly solving $F(x) = y$ near zero involves a function whose logarithmic derivative is large. In theorem 9.5, the large variable L is solved from a nearly quadratic, monotone equation; the final $x = a\mathcal{A}(L)e^{-L}$ is then formed without subtracting nearly equal quantities. The periodic series converges exponentially and usually requires only one harmonic in the dyadic case.

11 Legendre coefficients and approximation-theoretic implications

Let h_b be the density in (51), affinely rescaled to $[-1, 1]$ when necessary, and write its Legendre expansion

$$h_b(x) = \sum_{n \geq 0} a_n^{(b)} P_n(x), \quad a_n^{(b)} = \frac{2n+1}{2} \int_{-1}^1 h_b(x) P_n(x) dx. \quad (75)$$

Symmetry forces $a_{2n+1}^{(b)} = 0$. Standard repeated-integration estimates for Legendre coefficients, combined with theorem 7.3, allow the number of integrations to grow with the degree.

Proposition 11.1 (Log-squared upper envelope). *For each $b \geq 2$ there is $C_b > 0$ such that, along the even degrees,*

$$\log |a_{2n}^{(b)}| \leq -\frac{(\log n)^2}{2 \log b} + C_b \log n \log \log(n+2) \quad (n \geq 2). \quad (76)$$

The same scale applies to best polynomial approximation errors in every fixed Sobolev norm.

Proof sketch. A k -fold Legendre integration-by-parts estimate has the form

$$|a_n| \leq C^k n^{-k+O(1)} \max_{j \leq k} \|h_b^{(j)}\|;$$

the exact polynomial factors in n do not affect the leading log-squared term. By theorem 7.3, the derivative contribution is $\exp((\log b)k^2/2 + O_b(k))$. Choosing $k = \lfloor \log n / \log b \rfloor$ yields (76); tracking the variable- k constants in the Legendre estimate gives the displayed secondary term. $\square$

Conjecture 11.2 (Sharp Legendre envelope). *For every integer $b \geq 2$,*

$$\limsup_{n \rightarrow \infty} \frac{\log |a_{2n}^{(b)}|}{(\log n)^2} = -\frac{1}{2 \log b}. \quad (77)$$

More precisely, after removal of the quadratic and linear terms in $\log n$, the upper envelope contains a bounded nonconstant periodic function of $\log n / \log b$.

The conjectured constant is the Legendre-dual of the derivative-growth constant: optimizing $n^{-k} b^{k^2/2}$ produces exactly $\exp[-(\log n)^2/(2 \log b)]$. A lower bound is harder because Legendre oscillations can create cancellations absent from the Fourier resonant sequence. The supplied code computes exact rational moments for rational q , forms Legendre coefficients without quadrature, and plots the renormalized envelope.

12 Further conjectures and research programs

12.1 Exponentially improved endpoint transseries

Conjecture 12.1 (Geometric exponential transseries). *After the full periodic saddle series in powers of $1/L$ is removed, the endpoint remainder admits a transseries generated by exponentials*

$$e^{-zq^j}, \quad j = 0, 1, 2, \dots,$$

where $z = e^L$ is the saddle variable. The sectors accumulate geometrically and obey a q -difference resurgence law inherited from $H_q(z) = ((1 - e^{-z})/z)H_q(qz)$.

The superalgebraic remainder in theorem 8.2 is consistent with this claim but does not identify its exponentially small structure. A useful first goal is to derive the leading e^{-z} correction uniformly through one Mellin period and then track its descendants under $z \mapsto qz$.

12.2 Minor-arc closure of the saddle theorem

Problem 12.2. *Give a self-contained proof of theorem 9.1(c) with an explicit constant uniform for q in compact subsets of $(0, 1)$. Then compute the first three relative saddle coefficients and certify them by interval arithmetic.*

A direct route is to split the tilted product into active indices $k \leq L/r + O(1)$ and a small-argument tail. On the active block, a positive proportion of factors stays uniformly away from its local lattice of maxima; on the tail, the quadratic characteristic-function estimate is uniform. This is analogous to the minor-arc component of local limit theorems for digital harmonic sums.

12.3 Zero divisors beyond reciprocal integers

For general q , the multiplicity function is

$$m_q(z) = \#\{k \geq 0 : zq^k/\pi \in \mathbb{Z} \setminus \{0\}\}.$$

When $q^d = A/B$ is rational in lowest terms, the layers split into d residue classes, each carrying a mixed A – B valuation structure.

Problem 12.3. *Classify the canonical product and spectral Dirichlet series for all q such that some q^d is rational. Determine whether the resulting Dirichlet series factor into Riemann zeta functions and finitely many local Euler corrections, or whether genuinely new automatic Dirichlet series occur when the numerator $A > 1$.*

12.4 Geometric Bernoulli zeros and operators

The averaging operator

$$(T_q p)(x) = \mathbb{E}p(x + V_q)$$

has inverse on polynomials given by $T_q^{-1}x^n = \text{Ber}_{n,q}^{\text{geom}}(x)$. The fixed-point decomposition factors it as an infinite product of uniform-averaging operators.

Problem 12.4. *Develop the spectral and zero geometry of $\text{Ber}_{n,q}^{\text{geom}}$. Questions include the location of nonreal zeros, interlacing under $n \mapsto n+1$, asymptotic zero measures as $n \rightarrow \infty$, and operator factorizations of T_q^{-1} in terms of the q -difference law (C).*

Conjecture 12.5 (Central zero rigidity). *For each fixed $q \in (0, 1)$, all real zeros of $\text{Ber}_{n,q}^{\text{geom}}$ are simple except for the forced central zero of odd degree, and the number of real zeros is eventually a fixed proportion of n depending continuously on q .*

This is deliberately stated as a numerical research question rather than a theorem; classical Bernoulli polynomial zeros already warn against naive all-real-rootedness.

12.5 Strict log-concavity and total positivity

Conjecture 12.6 (Strict interior log-concavity). *For every $q \in (0, 1)$, $(\log u_q)''(x) < 0$ for $-1 < x < 1$, except possibly at the symmetry point in a limiting interpretation.*

Finite convolutions are spline densities for which strictness can be analyzed by variation-diminishing methods. Passing a uniform strict estimate to the infinite limit would give strong inequalities for F_q , including monotonicity of endpoint hazard ratios used in rigorous inverse bounds.

12.6 Formalization targets

Several results are well suited for formal proof assistants:

- (i) the probabilistic fixed point and functional equations;
- (ii) the affine-word induction in theorem 3.1;
- (iii) the Bell-polynomial formula (19);
- (iv) the moment recurrence and geometric Bernoulli inversion;
- (v) the valuation Hadamard product at the level of formal power series; and
- (vi) the upper half of the derivative-growth theorem.

The resonant lower bound additionally needs a modest amount of real/complex analysis. The Mellin contour argument is the least immediate formalization target but can be split into reusable theorems on harmonic sums.

13 Numerical experiments and reproducibility

The archive contains `experiments.py`. It uses arbitrary-precision arithmetic for products and Mellin coefficients, exact rational arithmetic where feasible, and standard floating-point quadrature only for the optional density plots. Running

```
python experiments.py -all
```

regenerates the CSV tables and PDF figures used here.

The experiments are designed as tests of identities, not as substitutes for proof:

- (1) `mellin_comparison.csv` compares the direct product $\log H_q(z)$ with (61), both with and without the periodic term.
- (2) `resonant_fourier.csv` checks (53) at high precision for several bases and levels.
- (3) `edgeworth_comparison.csv` compares numerical Fourier inversion with (42).
- (4) `legendre_coefficients.csv` computes dyadic even Legendre coefficients from exact moments, avoiding endpoint quadrature.
- (5) `corpus-manifest.txt` records every audited TeX source and its hash.

14 Synthesis

The geometric uniform convolution turns several themes that appear separately in the classical theory into facets of a single object.

- The two-branch differential refinement generates Thue–Morse signs; the branch locations carry a geometric q -deformation, and their signed moments are controlled by Bell polynomials with Bernoulli-number inputs.
- The infinite version of the same geometric power sums gives the exact cumulants. The reciprocal moment-generating function then produces an Appell family that extends classical Bernoulli polynomials and exactly inverts Fabius averaging.
- At reciprocal integer q , the Fourier zeros merge according to a digital valuation. The same base b fixes both the zero multiplicities and the sharp $n^2 \log b/2$ derivative-growth scale.
- Mellin poles at an imaginary arithmetic progression record discrete scale invariance. Their log-periodic term survives the saddle and supplies the missing phase in a genuinely complete endpoint expansion.
- Lambert W_{-1} remains the correct coarse inverse coordinate, but the Mellin phase specifies how to pass from a highly accurate approximation to an asymptotic calculus that is complete at fixed q .

The strongest concrete result is theorem 7.3: for the classical up-function, the exact leading logarithmic growth of every L^p derivative norm is $(\log 2)n^2/2$. The most useful new computational object is the phase-aware inverse parametrization (70)–(71). The broadest new structural object is the geometric Bernoulli sequence, which may be developed independently as a q -Appell/Nørlund theory.

A Selected algebraic checks

A.1 Consistency of the zeta and Bernoulli expansions

Euler’s formula

$$\zeta(2m) = (-1)^{m+1} \frac{B_{2m}(2\pi)^{2m}}{2(2m)!}$$

turns (49) into the cumulant expansion obtained from (24) after the rescaling $z = at$. This independently checks all normalizing factors in the Fourier and probabilistic derivations.

A.2 The first nonvanishing geometric Prouhet moment

Near $t = 0$,

$$2^n \prod_{j=0}^{n-1} \sinh(tq^j) = 2^n t^n q^{n(n-1)/2} + O(t^{n+2}),$$

so differentiation at zero gives $S_{n,n} = 2^n n! q^{n(n-1)/2}$. Substituting this into the Taylor expansion of (14) confirms that the normalization reproduces the n th derivative rather than a scalar multiple.

A.3 Scaling invariance of the quadratic derivative constant

If $g(x) = c^{-1}f(x/c)$, then

$$\|g^{(n)}\|_p = c^{-n-1+1/p} \|f^{(n)}\|_p.$$

The logarithm of the scaling factor is linear in n and therefore disappears after division by n^2 . This justifies moving freely between h_b and the support-normalized $u_{1/b}$ in theorem 7.3.

B Corpus manifest

The full manifest is supplied as a separate text file in the archive to keep this PDF readable. A compact generated summary is included below when available.

Corpus summary was not generated in this build. See `corpus-manifest.txt`.

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